Inferences on Proportions:

1) Single proportion:

on p.

2) Two proportions (diff of proportions)

X_1 & X_2

p_1 & p_2

Theory part for single proportion.

Random Variable: X

$X \sim \text{binom}(n, p)$

We will know 'n'.

p = prob of succes

= ?

p = population prop.

↳ fixed

P.E : sample proportion

$= \hat{p}$ ↳ change from sample to sample.

$\hat{p}$: random variable that models the sample proportion.

What is the sampling dist of $\hat{p}$?

$$E(\hat{p}) = p$$

$E(\hat{p})$ ↓ , s.e ↓

$$\hat{p} \sim N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$$

sampling dist.

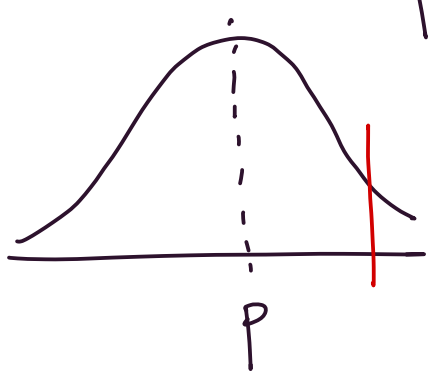
$$\hat{p} = \frac{1}{n} \cdot \text{number of successes.}$$

$$= \frac{1}{n} \varepsilon 1_s$$

similar to averages

$$\downarrow S.e = \sqrt{\frac{p(1-p)}{n}} \uparrow$$

$$\hat{p} \sim N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$$



$\uparrow C$

$\uparrow C.V$

CI : 100c% ci for p.

p.e $\pm$ (c.v x s.e) — margin of error.

$$\hat{p} \pm c.v \times \sqrt{\frac{p(1-p)}{n}}$$

s.e

Find out a way to know or estimate s.e:

Wald adjustment:

replace p with $\hat{p}$ in s.e

$$\hat{s.e} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$\downarrow$

s.e adj

CI of p :

$$\hat{p} \pm \underbrace{C.V}_{z} \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$z = qnorm\left(C + \frac{(1-C)}{2}\right) = qnorm\left(1 - \frac{\alpha}{2}\right)$$

$$C.V = qnorm\left(C + \frac{(1-C)}{2}\right)$$

↳ z-score.

(generally, $CV = qdist(-)$ ↓ samp. dist.

$$\hat{p} \pm \underbrace{qnorm\left(C + \frac{(1-C)}{2}\right)}_{1-\frac{\alpha}{2}} \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Inference on proportions:

1) Model: $p = ?$

$X \sim \text{binom}(n, p)$

(satisfies the BINS assump.

n = no. of trials (known)

p = prob of succ / pop - prop.

(unknown).

P.E: sample prop. $= \hat{p}$

Sampling dist of $\hat{p}$:

$$\hat{p} \sim N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$$

s.d of Binom r.v = $\sqrt{np(1-p)}$

$E(\hat{p}) = p$ & $s.e(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$ (unknown)

estimated $s.e(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

↑

Wald adj

AC adj $\doteq$ Agresti - Coull

when, values of p are on extreme ends of $[0,1]$.

C.I for p :

General formula: p.e ± c.v x s.e

$$\hat{p} \pm c.v \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

tail

$$z = c.v = qnorm\left(1 - \frac{\alpha}{2}\right)$$

$$c + \frac{1-c}{2}$$

$$\hat{p} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Hypothesis testing:

1) Model statement & stating the hypo.

Model: $X \sim \text{binom}(n, p)$

BINS assumptions.

State the hypotheses:

H_0: $p = $ value

H_a: $p \begin{matrix} > \\ < \\ \neq \end{matrix}$ value

Ex)

Muriel: $n = 8$

X = no. of cups she guesses correctly

$\sim \text{Binom}(8, p)$

Observed value of X : $x = 6$ (8)

H_0: $p = 0.5$ (guessing at random)

H_a: $p > 0.5$ (guessing with certainty).

If under the assumptions of hypo. testing, H_0 is true.

$$X_{H_0} \sim \text{Binom}(8, 0.5)$$

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p-val = Prob of getting $X = 6$ or something more extreme.

$P(X \geq 6) = 1 - pbinom(5, 8, 0.5)$

$= 0.144$

$\alpha = 0.05$

Conclusion: $p\text{-val} > \alpha$

$\Rightarrow$ We fail to reject H_0.

$P(X \geq 8) = 1 - pbinom(7, 8, 0.5) = 0.0039 < 0.05$

$\Rightarrow$ We reject H_0.

Hypothesis testing for single prop. under the binomial model.

X = no. of prosocial choice chimp A makes.

BINS.

$X \sim \text{binom}(90, p)$

p = prop of prosocial choices.

95% CI for p:

$$p.e = \hat{p} = \frac{x}{n}$$

x = no. of success = 60

$n = 90$

$$\hat{p} = \frac{60}{90}$$

$$\hat{p} \pm qnorm(0.975) \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\downarrow$$

$$0.667 \pm 1.96 \times 0.049$$

Inference on single proportion

sampling dist of $\hat{p}$

100(1-α)% CI for p : $\hat{p} \pm qnorm\left(1-\frac{\alpha}{2}\right) \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

adj s.e

p.e ± c.v × s.e

↳ depends on the p.e dist

Hypothesis testing for p :

1) Model statement & state the hypo :

H_0 : p = value

H_a : $p \begin{matrix} < \\ > \\ \neq \end{matrix}$ value

Binomial model: p.e = no. of success. = x

$\Rightarrow X \sim binom(n, p_{H_0})$

↳ complete model

p-value = pbinom(x_{obs}, n, p_{H_0})

or 1 - "

or pbinom(") +

(1 - pbinom(X_{obs}, n, p_{H_0}))

Look at the Psychic Example in the inference to single proportion section

Normal model : p.e : $\hat{p}$

$\sim N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$

find a test statistic :

$$z\text{-ts} = \frac{\hat{p} - p_{H_0}}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \sim N(0,1)$$

z test

value : pnorm (z , lower.tail =)

or 2 × "

Inference on diff in prop.

$X_1 \sim \text{Binom}(n_1, p_1)$ | $X_2 \sim \text{Binom}(n_2, p_2)$

$\downarrow$ | $\downarrow$

p_1 | p_2

$p_1 - p_2$ $\quad = + \rightarrow p_1$ is bigger

$\quad - \rightarrow p_1$ is smaller

individual p.e: $\hat{p}_1 \sim N\left(p_1, \sqrt{\frac{p_1(1-p_1)}{n_1}}\right)$ & $\hat{p}_2 \sim N\left(p_2, \sqrt{\frac{p_2(1-p_2)}{n_2}}\right)$

(sample proportion from X_1)

(sample prop. from X_2)

parameter : $p_1 - p_2$

p.e : $\hat{p}_1 - \hat{p}_2$

$E(\text{p.e}) = \text{parameter}.$

$$E(\hat{p}_1 - \hat{p}_2) = E(\hat{p}_1) - E(\hat{p}_2)$$
$$= p_1 - p_2$$

$$\hat{p}_1 - \hat{p}_2 \sim N\left(p_1 - p_2, \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}\right)$$

$$\left(\begin{array}{l} a \sim N(\mu_a, \sigma_a) \\ b \sim N(\mu_b, \sigma_b) \end{array}\right) = a + b \sim N\left(\mu_a + \mu_b, \sqrt{\sigma_a^2 + \sigma_b^2}\right)$$

$$E(\hat{p}_1 - \hat{p}_2) = p_1 - p_2$$ → center of samp. dist.

$$s.e(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

↳ s.e of samp. dist.

Wald adj. s.e:

$$\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n}}$$

C.I for $p_1 - p_2$:

$$(\hat{p}_1 - \hat{p}_2) \pm qnorm\left(c + \frac{1-c}{2}\right) \times \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

p.e C.V S.e

Hypothesis testing for $p_1 - p_2$

p.e: $\hat{p}_1 - \hat{p}_2$ parameter: $p_1 - p_2$

Sampling dist of p.e:

$$N\left(p_1 - p_2, \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}\right)$$

H_0 is assumed to be true

$H_0: p_1 - p_2 = 0$ (that both proportions are the same)

$H_a: p_1 - p_2 \begin{matrix} > \\ < \\ \neq \end{matrix} 0$

1) Model statement & state the hypo.

Under H_0, $p_1 = p_2 = p$

↳ global proportion for the pop.

Sampling dist under H_0:

$$N\left(p_1 - p_2, \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}\right)$$

$$\downarrow$$

$$N\left(0, \sqrt{\frac{p(1-p)}{n_1} + \frac{p(1-p)}{n_2}}\right)$$

$$s.e = \sqrt{\frac{p(1-p)}{n_1} + \frac{p(1-p)}{n_2}}$$

Global proportion: $\tilde{p} = \frac{x_1 + x_2}{n_1 + n_2}$

$$s.e_{adj} = \sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}$$

2) test statistic & null dist.

$$ts \overset{z}{=} \frac{(\hat{p}_1-\hat{p}_2) - (p_1-p_2)_{H_0 \to 0}}{s.e_{adj}(\hat{p}_1-\hat{p}_2)}$$

$$= \frac{(\hat{p}_1-\hat{p}_2) - 0}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}} \sim N(0,1)$$

null dist.

3) p-val.

p-val = pnorm(−2) (left tailed)

= pnorm(z, lower.tail = F) (right-tailed)

2 * pnorm(|z|, lower.tail = F)

(

p-val = pnorm(|z|, lower.tail = F)

4) conclusion

p-val with α.

p-val $< \alpha \rightarrow$ Reject H_0

p-val $\geq \alpha \rightarrow$ We fail to reject H_0.

Inference on single mean.

CLT: $X \sim D(\mu, \sigma)$ — $E(X) = \mu$, $sd(X) = \sigma$

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

$E(\bar{X}) = \mu$

Model: Considering any rv X.

with some dist D.

& some $E(X) = \mu$ & $sd(X) = \sigma$

$$X \sim D(\mu, \sigma)$$

↑ parameter of interest

unknown: D, μ, σ